

The Essay Plan Roll No. 530

Topic: Are humans still evolving by natural selection?

This essay plan will show the main points of the essay on whether humans are still evolving by natural selection and the basic methods for the essay.

I believe that humans are still evolving by natural selection. Here are two helpful dimensions for me to find out the proof of human evolution by natural selection: 1) the adaptive evolution which results from natural selection; 2) the understanding of the concepts of natural selection and evolution.

- 1) For the first dimension, as is said in *Biology: A Global Approach*, natural selection is the only mechanism that leads to adaptive evolution consistently. (Campbell, et al., 2016) If this conclusion is true, evidence for adaptive evolution within people can be evidence for human evolution by natural selection. In this portion different examples will be used to examine the idea.
 - a) For example, a study led by John Hawks indicates that the adaptive evolution of human is still ongoing, experiencing an acceleration of positively selected alleles. (Hawks, et al., 2007). This result may also indicate natural selection. However, the authors admit that the rate of acceleration is lower than expected, which is believed to be a result of subtler natural selection and other reasons, (Hawks, et al., 2007) which requires further research.
 - b) Nevertheless, some people argued that adaptive evolution does not necessarily result from natural selection. (Kull, 2014) Other mechanisms are believed to cause adaptive evolution. For example, grasshoppers living in a pavement differ in color maybe because of the imprinting effect and matching habitat choice, even if there is no natural selection. (Edelaar, et al., 2015) This decreases the credibility of the method used here. Therefore, more research should be done about the credibility of the hypothesis above and if the hypothesis is true, then examine whether imprinting and matching habitat choice result from natural selection.
- 2) For the understanding of natural selection, a thorough understanding of the term – “relative fitness” is required. Note that natural selection happens not always with the fierce contest for survival. The way to get reproductive success is subtler and depends on many factors. (Campbell, et al., 2016) In this portion, a discussion over the terms and their meaning will be made.
 - a) Competition for reproductive success is key to natural selection and can take place in various subtle ways, instead of struggle for survival alone. For example, social competition may also serve as a method of getting reproductive success. (West-Eberhard, 1979) This indicates the selection within humans that leads to evolution.

In summary, further research should be done to check the validity of the evidence for both human evolution by natural selection and the alternative hypothesis before reaching a solid conclusion.

Bibliography

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